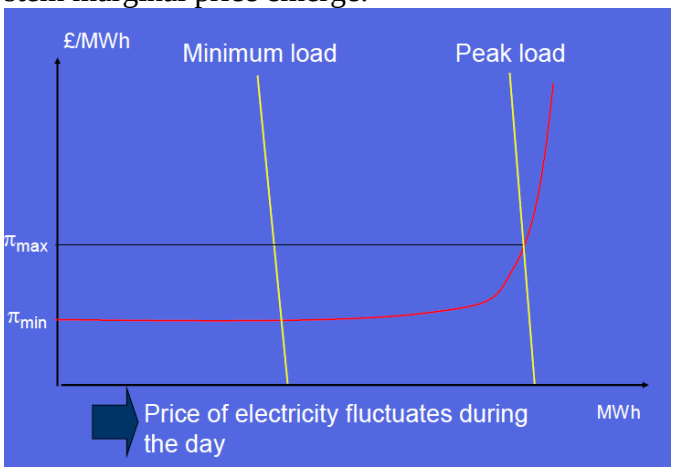
Solution to Question 1

(a) The system marginal price signifies the marginal cost of serving an additional unit of demand at the least-cost dispatch of the total load to the available generators. In contrast with other commodities, electrical energy cannot be economically stored in large quantities and thus it must be produced exactly when it is consumed by the inflexible demand. Moreover, the electrical demand is not the same every hour, day, month etc and the generation side is characterized by a production cost which varies with the size of the load. The combination of these effects explains why the system marginal price varies with time. This is demonstrated in the figure below. The supply curve of the generation side exhibits a "hockey-stick" shape, being rather flat at the area of low loads, where cheap base generators are employed for the satisfaction of the demand, and rather steep at the area of high loads, where expensive peaking generators are also employed to cover the demand. In combination with the fluctuation of the demand from a minimum to a peak value, temporal variations in the system marginal price emerge.



[2 marks]

(b) In the optimal solution of the problem, the marginal cost of each generator is equal to the system marginal cost λ yielding:

$$\frac{\partial C_1(P_1)}{\partial P_1} = 2 + 0.1 \cdot P_1 = \lambda \rightarrow P_1 = \frac{\lambda - 2}{0.1} \quad (1)$$

$$\frac{\partial C_2(P_2)}{\partial P_2} = 8 + 0.5 \cdot P_2 = \lambda \rightarrow P_2 = \frac{\lambda - 8}{0.5} \quad (2)$$

(1.2)

$$\frac{\partial C_3(P_3)}{\partial P_3} = 16 + 1.1 \cdot P_3 = \lambda \rightarrow P_3 = \frac{\lambda - 16}{1.1} \quad (3)$$

The satisfaction of the system demand implies that:

$$P_1 + P_2 + P_3 = D = 600\ [MW] \quad (4)$$

Substituting (1)-(3) in (4) gives for the system marginal cost:

$$\frac{\lambda - 2}{0.1} + \frac{\lambda - 8}{0.5} + \frac{\lambda - 16}{1.1} = 600 \rightarrow \lambda = 50.39\ [\frac{£}{MWh}] \quad (5)$$

Substituting (5) in (1)-(3) yields for the optimal production of each generator:

$P_1 = 483.94\ [MW]$ (6)

$P_2 = 84.79\ [MW]$ (7)

$P_3 = 31.27\ [MW]$ (8)

[2 marks]

Substituting (6)-(8) in the cost functions of the generators yields for the total cost of operating the system:

$C_{tot} = C_1(P_1) + C_2(P_2) + C_3(P_3) = 18141.55\ [£]$ (9)

The system average cost is:

$\frac{C_{tot}}{P_1+P_2+P_3} = 30.24\ [\frac{£}{MWh}]$ (10)

[2 marks]

The profit for each generator Ω_i is given by the difference between its respective revenue and cost:

$\Omega_1 = \lambda \cdot P_1 - C_1(P_1) = 10910.07\ [£]$ (11)

$\Omega_2 = \lambda \cdot P_2 - C_2(P_2) = 1147.28\ [£]$ (12)

$\Omega_3 = \lambda \cdot P_3 - C_3(P_3) = 37.71\ [£]$ (13)

[2 marks]

(c) It would produce up to the point where the marginal cost of each unit is 42.60 £/MWh. The rest will be bought from the market at 42.60 £/MWh.

$2 + 0.1 \cdot P_1 = 42.60 \rightarrow P_1 = \frac{42.60-2}{0.1} = 406.00\ [MW]$ (14)

$8 + 0.5 \cdot P_2 = 42.60 \rightarrow P_2 = \frac{42.60-8}{0.5} = 69.20\ [MW]$ (15)

$16 + 1.1 \cdot P_3 = 42.60 \rightarrow P_3 = \frac{42.60-16}{1.1} = 24.18\ [MW]$ (16)

[2 marks]

So the rest 600 – 406.00 – 69.20 – 24.18 = 100.62 MW will be bought from the spot market.

[1 mark]

(d) Given that $P_3 < P_3^{min}$ the solution calculated in **(b)** is not feasible when taking into account given the minimum output limits of the generators. The optimal solution can now be calculated by fixing the output of generator 3 to:

$P_3' = P_3^{min} = 40\ [MW]$ (17)

and satisfying the remaining demand:

$D' = D - P_3' = 560\ [MW]$ (18)

through the remaining generators 1 and 2. Following the same method as in (b), we get:

$\frac{\lambda'-2}{0.1} + \frac{\lambda'-8}{0.5} = 560 \rightarrow \lambda' = 49.67\ [\frac{£}{MWh}]$ (19)

Substituting (19) in (1)-(2) yields for the optimal production of each generator:

$P_1' = 476.67[MW]$ (20)

$P_2' = 83.33\ [MW]$ (21)

[1 mark]

Substituting (17) and (20)-(21) in the cost functions of the generators yields for the total cost of operating the system:

$C_{tot}' = C_1(P_1') + C_2(P_2') + C_3(P_3') = 18186.67\ [£]$ (22)

[1 mark]

The system average cost is:

$$\frac{C'_{tot}}{P'_1+P'_2+P'_3} = 30.31\ [\frac{£}{MWh}] \quad (23)$$

[1 mark]

The profit for each generator Ω_i is given by the difference between its respective revenue and cost:

$$\Omega'_1 = \lambda \cdot P'_1 - C_1(P'_1) = 10560.56\ [£] \quad (24)$$

$$\Omega'_2 = \lambda \cdot P'_2 - C_2(P'_2) = 1086.11\ [£] \quad (25)$$

$$\Omega'_3 = \lambda \cdot P'_3 - C_3(P'_3) = -33.33\ [£] \quad (26)$$

[2 marks]

When the maximum output limits are considered, the output of generator 3 needs to be restricted to a higher value and therefore the outputs of the other two generators are reduced. Given that generator 3 is characterised by the highest variable cost, this new dispatch is more costly (i.e. it increases the total cost of operating the system, the system marginal cost and the average system cost). On the other hand, this new dispatch reduces the profit of all the generators. For generators 3 this is driven by the reduced system marginal cost and the increased generation output, while for generators 1-2 this is the reduction of both their generation outputs and the system marginal cost.

[1 mark]

(e) The negative profit of generator 3 means that the operating cost of this generator is larger than its revenue in the optimal solution and thus it experiences economic losses at the optimal solution (does not recover its operating costs). This is happening since marginal system cost pricing does not guarantee the recovery of the fixed (no-load) costs of the participating generators in combination with the high fixed costs of generator 3. This lack of recovery of some of the generators' fixed costs will mean in the long term that these generators will abandon the business resulting potentially in higher total costs of operating the system or even security problems (inability of the system to satisfy the demand). Therefore, a suitable mechanism compensating the generators' unrecovered fixed costs is required in top of the marginal system cost pricing.

[3 marks]

Solution to Question 2:

(a) Unbundling refers to the process of separating the activities of generation, transmission, distribution and retail of electricity, previously performed by a single vertically integrated utility, into different entities i.e. companies. This then allows for the competition to be introduced in the generation and retail business, while transmission and distribution are organised as regulated monopolies, ensuring fair and open network access to all market participants. The reasons for liberalising electricity markets are the increased efficiency in the supply of electricity and the lower cost of electricity to consumers. This is achieved through introducing competition both at the wholesale level (generators compete to sell electrical energy) and the retail level (consumers choose from whom they buy electricity).
[2 marks]
If a generator bids higher than its marginal cost, it could become an extra-marginal producer and miss an opportunity to sell at a profit. If the generator bids lower than its marginal cost, it could have produced at a loss. Any of these two options bring in a certain amount of risk to the producer that would rather be avoided. On the other hand, if the producer bids at its marginal cost, it gets paid the market price if it is a marginal or infra-marginal producer. This is the preferred option for a producer.
[2 marks]

(b) Under perfect competition, the price is equal to the marginal cost of each generator. Therefore, the following two equations hold:

$$\frac{\partial C_A(P_A)}{\partial P_A} = \pi \quad (1)$$

$$\frac{\partial C_B(P_B)}{\partial P_B} = \pi \quad (2)$$

Since $\pi = 125 - 1.7 \cdot D$ and $D = P_A + P_B$ (demand-supply balance condition), equations (1) and (2) yield respectively:

$$35 + 0.74 \cdot P_A = 125 - 1.7 \cdot (P_A + P_B) \rightarrow P_A = 36.8852 - 0.6967 \cdot P_B \quad (3)$$

$$39 + 0.5 \cdot P_B = 125 - 1.7 \cdot (P_A + P_B) \rightarrow P_B = 39.0909 - 0.7727 \cdot P_A \quad (4)$$

The combination of (3) and (4) yields:

$$P_A = 20.9040\ [MW] \quad (5)$$

$$P_B = 22.9379\ [MW] \quad (6)$$

[1 mark]
The total demand is given by:

$$D = P_A + P_B \rightarrow D = 43.8418\ [MW] \quad (7)$$

[1 mark]
The electricity price is given by:

$$\pi = 125 - 1.7 \cdot D \rightarrow \pi = 50.4689\ [\frac{£}{MWh}] \quad (8)$$

[1 mark]
The profits made by the two generators are given by:

$$\Omega_A = P_A \cdot \pi - 35 \cdot P_A - 0.37 \cdot P_A^2 \rightarrow \Omega_A = 161.6809\ [\frac{£}{h}] \quad (9)$$

$$\Omega_B = P_B \cdot \pi - 39 \cdot P_B - 0.25 \cdot P_B^2 \rightarrow \Omega_B = 131.5363 \left[\frac{£}{h}\right] \quad (10)$$

[1 mark]

(c) (i) In the Cournot model of competition the state of the market is determined by the production decisions made by each firm. We summarize the possible outcomes using a table where all the cells in a column correspond to a given production by company A and the cells in a row correspond to a given production by company B.
Each cell contains four pieces of information arranged in the following format:

D	Ω_A
Ω_B	π

where:

π price $\left[\frac{£}{MWh}\right]$

D demand $[MW]$

Ω_A profit made by company A $\left[\frac{£}{h}\right]$

Ω_B profit made by company B $\left[\frac{£}{h}\right]$

Given the productions P_A and P_B of the two companies, the other quantities are calculated as follows:

$$D = P_A + P_B \quad (11)$$

$$\pi = 125 - 1.7 \cdot D \quad (12)$$

$$\Omega_A = P_A \cdot \pi - 35 \cdot P_A - 0.37 \cdot P_A^2 \quad (13)$$

$$\Omega_B = P_B \cdot \pi - 39 \cdot P_B - 0.25 \cdot P_B^2 \quad (14)$$

Based on the above, the table expressing the Cournot model of competition for the conditions of the problem is shown below.

P_B/P_A	14		15		16	
15	29	497.28	30	501.75	31	502.08
	494.25	75.7	468.75	74.0	443.25	72.3
16	30	473.48	31	476.25	32	474.88
	496.00	74.0	468.80	72.3	441.60	70.6
17	31	449.68	32	450.75	33	447.68
	493.85	72.3	464.95	70.6	436.05	68.9

[6 marks]

A Nash equilibrium state implies that given the production level of either of the two companies in this state, the other company cannot achieve a higher profit by following a

production level different than the one at the equilibrium. It can thus be observed form the table below that the cell corresponding to $P_A = 16MW$ and $P_B = 15MW$; $P_A = 15MW$ and $P_B = 16MW$ are two Nash equilibria. In other words, given that $P_A = 16MW$, company B achieves its highest profit by producing $P_B = 15MW$ AND given that $P_B = 15MW$, company A achieves its highest profit by producing $P_A = 16MW$. Furthermore, given that $P_A = 15MW$, company B achieves its highest profit by producing $P_B = 16MW$ AND given that $P_B = 16MW$, company A achieves its highest profit by producing $P_A = 15MW$.
[4 marks]

(c) (ii) We can observe that under imperfect competition the market price and the generation profits are significantly higher than under perfect competition, since producers exercise market power and manipulate the prices to strategically increase their profits. As a result of the price elasticity of the demand, the increased price leads to lower demand (and consequently to lower generation levels).
[2 marks]

Solutions to Question 3

(a) Since the generators' marginal costs are fixed, the minimum cost dispatch can be calculated by dispatching the generators in an ascending marginal cost order until the total demand of $D_1 + D_2 + D_3 = 500\ [MW]$ is satisfied. We thus get for the minimum cost dispatch:

$P_a = 155\ [MW]$ (1)

$P_b = 345\ [MW]$ (2)

$P_c = 0\ [MW]$ (3)

$P_d = 0\ [MW]$ (4)

[2 marks]

The line flow problem can be solved using the superposition principle and also directly. To this effect, we write the power balance equation at two buses and KVL around the loop.

Bus 1: $P_a + P_b - 80 = F_{12} + F_{13}$ (5)

Bus 2: $P_c - 90 = -F_{12} + F_{23}$ (6)

Bus 3: $P_d - 330 = -F_{13} - F_{23}$ (7)

Loop equation: $0.3 \cdot F_{12} + 0.2 \cdot F_{23} - 0.3 \cdot F_{13} = 0$ (8)

Putting these equations in the matrix form gives:

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & -1 \\ 0.3 & -0.3 & 0.2 \end{bmatrix} \begin{bmatrix} F_{12} \\ F_{13} \\ F_{23} \end{bmatrix} = \begin{bmatrix} P_c - 90 \\ P_d - 330 \\ 0 \end{bmatrix} \quad (9)$$

Substituting (3.1)-(3.4) in (3.9), we get:

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & -1 \\ 0.3 & -0.3 & 0.2 \end{bmatrix} \begin{bmatrix} F_{12} \\ F_{13} \\ F_{23} \end{bmatrix} = \begin{bmatrix} -90 \\ -330 \\ 0 \end{bmatrix} \quad (10)$$

Solving these equations, we get:

$F_{12} = 180\ [MW]$ (11)

$F_{13} = 240\ [MW]$ (12)

$F_{23} = 90[MW]$ (13)

Since $F_{12} > F_{12}^{max}$ ($F_{12}^{max} = 160\ [MW]$ according to the data) the branch 1-2 is overloaded.

[3 marks]

(b) In order to eliminate this overload the output of generator G_d is increased and the output of generator G_a is decreased by the same amount. Decreasing the output of generator G_b is not desirable as it is cheaper than generator G_a.

To calculate how big this increase should be to remove the violation of the flow limit on line 1-2, consider an injection of $+1[MW]$ at bus 3 and an injection of $-1[MW]$ at bus 1. This pair of injection causes a flow in the network that divides itself as follows:

$\frac{0.3}{(0.3+0.2)+0.3} \cdot 1 = 0.375[MW]$ along the path 3-2-1 (14)

$\frac{(0.3+0.2)}{(0.3+0.2)+0.3} \cdot 1 = 0.625[MW]$ along the path 3-1 (15)

Since we use a linear (dc) model, we can say that to remove $F_{12} - F_{12}^{max} = 20\ [MW]$ overload on line 1-2, we therefore need to increase the output of G_d by:

$$\frac{20}{0.375} = 53.33[MW] \quad (16)$$

Thus, the new generators' dispatch becomes:

$$P'_a = 101.67\ [MW] \quad (17)$$

$$P'_b = 345\ [MW] \quad (18)$$

$$P'_c = 0\ [MW] \quad (19)$$

$$P'_d = 53.33[MW] \quad (20)$$

[2 marks]

Using the nodal and loop equations, the flows are calculated solving the following linear system:

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & -1 \\ 0.32 & -0.32 & 0.17 \end{bmatrix} \begin{bmatrix} F'_{12} \\ F'_{13} \\ F'_{23} \end{bmatrix} = \begin{bmatrix} P'_a + P'_b - 80 \\ P'_d - 330 \\ 0 \end{bmatrix} \quad (21)$$

We get:

$$F'_{12} = 160\ [MW] \quad (22)$$

$$F'_{13} = 206.67\ [MW] \quad (23)$$

$$F'_{23} = 70[MW] \quad (24)$$

which satisfy the lines' capacities constraints.

The price at bus 2 is the cost of supplying an additional MW of load at this bus without violating the lines' capacities' constraints. The output of generator G_b cannot be increased since the latter runs at full output according to (18) and the given data. Increasing the output of generator G_a or generator G_d per 1 MW will lead to a violation of line 1-2 capacity constraint. Increasing the output of generator G_c will not lead to such a violation but seems as a costly alternative since in that case the price at bus 2 is:

$$\pi_2 = MC_c = 35\ [£/\text{MWh}] \quad (25)$$

Our goal is to check whether a combination of modifications in the outputs of generators G_a and G_d can lead to a lower price than the one in (25).

Extracting an additional $1[MW]$ at bus 2 and generating it at bus 1 with G_a causes the following change in the flow on line 1-2:

$$\begin{bmatrix} 1 & 1 & 0 \\ -1 & 0 & 1 \\ 0.3 & -0.3 & 0.2 \end{bmatrix} \begin{bmatrix} \Delta F_{12} \\ \Delta F_{13} \\ \Delta F_{23} \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} \Delta F_{12} \\ \Delta F_{13} \\ \Delta F_{23} \end{bmatrix} = \begin{bmatrix} 0.625 \\ 0.375 \\ -0.375 \end{bmatrix} \quad (26)$$

Similarly, extracting an additional $1[MW]$ at bus 2 and generating it at bus 3 with G_d causes the following change in the flow on line 1-2:

$$\begin{bmatrix} 0 & -1 & -1 \\ -1 & 0 & 1 \\ 0.3 & -0.3 & 0.2 \end{bmatrix} \begin{bmatrix} \Delta F_{12} \\ \Delta F_{13} \\ \Delta F_{23} \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} \Delta F_{12} \\ \Delta F_{13} \\ \Delta F_{23} \end{bmatrix} = \begin{bmatrix} 0.25 \\ -0.25 \\ -0.75 \end{bmatrix} \quad (27)$$

Therefore, increase of 1 MW of the output of G_a and G_d will increase the flow in line 1-2 by 0.625 MW and 0.25 MW respectively.

We need a total increase of 1 MW in the outputs of G_a and G_d (28) but without increasing the power flow in line 1-2 (29):

$$\Delta P_a + \Delta P_d = 1\ [MW] \quad (28)$$

$$0.625 \cdot \Delta P_a + 0.25 * \Delta P_d = 0 \quad (29)$$

which yields:

$\Delta P_a = -0.6667[MW]$ (30)
$\Delta P_d = 1.6667\ [MW]$ (31)
When these outputs' modifications are carried out the price at bus 2 is:
$\pi_2' = -0.6667 \cdot MC_A + 1.6667 \cdot MC_D = 29[£/\text{MWh}]$ (32)
[3 marks]
Which is accepted as the final value of price at bus 2 since it is lower than the value in (25).

(c) Although the price at bus 2 is 29 [£/MWh] (32), the contract for difference that Load at bus 2 has made at the price of 26.5 [£/MWh] means that it will pay for satisfying its demand:
$C_2 = 29 \cdot D_2 + (26.5 - 29) \cdot D_2 = 2385\ [£/\text{h}]$ (33)
Its income from the transmission congestion contract (regarding line 1-2) it has made is given by the product of the volume of the flow in the contract (90 $[MW]$) and the price differential in the two buses where line 1-2 is connected. In other words:
$I_2 = 90 * (\pi_2' - \pi_1)$ (34)
Since generator G_a is the cheapest generator, whose output has not reached its maximum limit (17)-(20), the price at bus 1 can be easily shown to be:
$\pi_1 = MC_a = 19\ [£/\text{MWh}]$ (35)
[1 mark]
Substituting (32) and (35) in (34) gives:
$I_2 = 900\ [£/\text{h}]$ (36)
Therefore, the net cost that load at bus 2 has to pay is (from (33) and (36)):
$C_2^{net} = C_2 - I_2 = 1485\ [£/\text{h}]$ (37)
[2 marks]
and the unit price that demand at bus 2 will pay is:
$\pi_2^{net} = \frac{C_2^{net}}{D_2} = 16.5\ [£/\text{MWh}]$ (38)
[2 marks]

(d) The flows calculated in **(b)** now lead to an overload of line 2-3. In order to eliminate this overload, the output of generator G_d is increased and the output of generator G_a is reduced by the same amount. Increase of 1 MW of the output of G_d has been shown in equation (14) to cause a 0.375 MW reduction in F_{23}. Therefore, in order to cause a reduction of $F_{23}' - F_{23}^{max} = 3\ [MW]$, we therefore need to increase the output of G_d by:
$\frac{3}{0.375} = 8\ [MW]$ (39)
Therefore the new dispatch is:
$P_a'' = 93.67\ [MW]$ (40)
$P_b'' = 345\ [MW]$ (41)
$P_c'' = 0\ [MW]$ (42)
$P_d'' = 61.33\ [MW]$ (43)
Using the nodal and loop equations, the flows are calculated solving the following linear system:
$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & -1 & -1 \\ 0.3 & -0.3 & 0.2 \end{bmatrix} \begin{bmatrix} F_{12}'' \\ F_{13}'' \\ F_{23}'' \end{bmatrix} = \begin{bmatrix} P_a'' + P_b'' - 80 \\ P_d'' - 330 \\ 0 \end{bmatrix} \quad (44)$$
the new network flows are:

$F''_{12} = 157\ [MW]$ (45)
$F''_{13} = 201.67\ [MW]$ (46)
$F''_{23} = 67\ [MW]$ (47)
Price at bus 2 is calculated by following the same methodology as in **(b)**. Increase of 1 MW of the output of G_a and G_d can be easily shown to reduce the flow in line 2-3 per 0.375 MW and 0.75 MW respectively, as shown in (26)-(27).
We need a total increase of 1 MW in the outputs of G_a and G_d (28) but without increasing the power flow in line 2-3 (29):
$\Delta P'_a + \Delta P'_d = 1\ [MW]$ (48)
$-0.375\Delta P'_a - 0.75 * \Delta P'_d = 0$ (49)
which yields:
$\Delta P'_a = 2\ [MW]$ (50)
$\Delta P'_d = -1\ [MW]$ (51)
When these outputs' modifications are carried out the flow in line 1-2 becomes (according to (26)-(27)):
$F'''_{12} = F''_{12} + \Delta P'_a \cdot 0.625 + \Delta P'_d \cdot 0.25 = 158\ [MW]$ (52)
and thus, capacity constraint of line 1-2 is not violated. The price at bus 2 is:
$\pi''_2 = 2 \cdot MC_A - 1 \cdot MC_D = 13\ [£/\text{MWh}]$ (53)
[2 marks]
Based on its contract for differences, load at bus 2 will now pay for satisfying its demand:
$C'_2 = 13 \cdot D_2 + (26.5 - 13) \cdot D_2 = 2385\ [£/\text{h}]$ (54)
Its income from the transmission congestion contract is now:
$I'_2 = 90 \cdot (\pi''_2 - \pi_1) = -540\ [£/\text{h}]$ (55)
Therefore, the net cost that load at bus 2 has to pay is (from (54) and (55)):
$C_2^{net\prime} = C'_2 - I'_2 = 2925\ [£/\text{h}]$ (56)
and the unit price that demand at bus 2 will pay is:
$\pi_2^{net\prime} = \frac{C_2^{net\prime}}{D_2} = 32.5\ [£/\text{MWh}]$ (57)
[3 marks]

Solution to Question 4

(a) We neglect the transmission line constraint.

Winter

To find the optimal levels of generation we need to solve the problem:

Minimize $C_A(P_A) + C_B(P_B)$ $subject\ to$ $P_A + P_B = 3500$

$$L = C_A(P_A) + C_B(P_B) - \lambda(P_A + P_B - 3500) \quad (1)$$

$$\frac{\partial L}{\partial P_A} = 0 \rightarrow P_A = \frac{\lambda - 1}{0.002} \quad (2)$$

$$\frac{\partial L}{\partial P_B} = 0 \rightarrow P_B = \frac{\lambda - 2}{0.004} \quad (3)$$

$$P_A + P_B = 3500 \rightarrow \lambda = 6\frac{\$}{MWh} = MC_A = MC_B, P_A = 2500MW, P_B = 1000MW \quad (4)$$

[2 marks]

Summer

To find the optimal levels of generation we need to solve the problem:

Minimize $C_A(P_A) + C_B(P_B)$ $subject\ to$ $P_A + P_B = 2000$

$$L = C_A(P_A) + C_B(P_B) - \lambda(P_A + P_B - 2000) \quad (5)$$

$$\frac{\partial L}{\partial P_A} = 0 \rightarrow P_A = \frac{\lambda - 1}{0.002} \quad (6)$$

$$\frac{\partial L}{\partial P_B} = 0 \rightarrow P_B = \frac{\lambda - 2}{0.004} \quad (7)$$

$$P_A + P_B = 2000 \rightarrow \lambda = 4\frac{\$}{MWh} = MC_A = MC_B, P_A = 1500MW, P_B = 500MW \quad (8)$$

[2 marks]

(b) We take into account the transmission line constraint of 1200MW.

Winter

The optimal dispatch becomes: $P_A = 1200MW,\ P_B = 2300MW$ and the prices are:

$$MC_A = 3.4\frac{£}{MWh} = PriceA \quad (9)$$

$$MC_B = 11.2\frac{£}{MWh} = PriceB \quad (10)$$

$$C_A = 2{,}640\frac{£}{h},\ C_B = 15{,}180\frac{£}{h} \quad (11)$$

[2 marks]

Summer

The optimal dispatch becomes: $P_A = 1200MW,\ P_B = 800MW$ and the prices are

$$MC_A = 3.4\frac{£}{MWh} = PriceA \quad (12)$$

$$MC_B = 5.2\frac{£}{MWh} = PriceB \quad (13)$$

$$C_A = 2{,}640\frac{£}{h},\ C_B = 2{,}880\frac{£}{h} \quad (14)$$

[2 marks]

(c) (i) We take into account the transmission line constraint of 1200MW.

The annual generation costs are:

$C_A^{winter} = 2{,}640\frac{£}{h} \cdot 2700h = 7{,}128{,}000\,£$ (15)

$C_B^{winter} = 15{,}180\frac{£}{h} \cdot 2700h = 40{,}986{,}000\,£$ (16)

$C_A^{summer} = 2{,}640\frac{£}{h} \cdot 6060h = 15{,}998{,}400\,£$ (17)

$C_B^{summer} = 2{,}880\frac{£}{h} \cdot 6060h = 17{,}452{,}800\,£$ (18)

$C_{A+B}^{annual} = 81{,}565{,}200\,£$ (19)

[2 marks]

(c) (ii) The electricity price in winter is 11.2 £/MWh and in summer is 5.2 £/MWh.
The annual demand charges are:

$3500MW \cdot 2700h \cdot 11.2\frac{£}{MWh} + 2000MW \cdot 6060h \cdot 5.2\frac{£}{MWh} = 168{,}864{,}000£$ (20)

The annual generation revenues are:

$R_A^{winter} = 1200MW \cdot 3.4\frac{£}{MWh} \cdot 2700h = 11{,}016{,}000\,£$ (21)

$R_B^{winter} = 2{,}300MW \cdot 11.2\frac{£}{MWh} \cdot 2700h = 69{,}552{,}000\,£$ (22)

$R_A^{summer} = 1200MW \cdot 3.4\frac{£}{MWh} \cdot 6060h = 24{,}724{,}800\,£$ (23)

$R_B^{summer} = 800MW \cdot 5.2\frac{£}{MWh} \cdot 6060h = 25{,}209{,}600\,£$ (24)

$R_{A+B}^{annual} = 130{,}502{,}400\,£$ (25)

[2 marks]

(c) (iii) The TransCo's total annual revenue is

$R_{Tra} = 1200MW(11.2 - 3.4) \cdot 2700h + 1200 \cdot (5.2 - 3.4) \cdot 6060 = 38{,}361{,}600£$ (26)

We can see that it is equal to the difference between annual demand charges and annual generation revenues.

[2 marks]

(d) After the reinforcement, the transmission line constraint becomes (1200+900) = 2100MW

Winter

The optimal dispatch becomes: $P_A = 2100MW,\ P_B = 1400MW$ and the prices are:

$MC_A = 5.2\frac{£}{MWh} = PriceA$ (27)

$MC_B = 7.6\frac{£}{MWh} = PriceB$ (28)

[1 mark]

Summer

The optimal dispatch becomes: $P_A = 1500MW,\ P_B = 500MW$ and the prices are

$MC_A = 4\frac{£}{MWh} = PriceA$ (29)

$MC_B = 4\frac{£}{MWh} = PriceB$ (30)

[1 mark]

Thus, Borduria TransCo's total annual revenue will be:

$R_{Tra} = 2100MW \cdot (7.6 - 5.2) \cdot 2700h = 13{,}608{,}000\,£$ (31)

The reinforcement will cost £5,000,000.

[1 mark]
Also, total annual demand charges:
$$3500 \cdot 2700 \cdot 7.6 + 2000 \cdot 6060 \cdot 4 = 120{,}300{,}000\ £ \quad (28)$$
i.e. a reduction of £48,564,000.
[1 mark]
In addition, the total annual generation costs will be
$$3750 \cdot 6060 + 1500 \cdot 6060 + 6510 \cdot 2700 + 6720 \cdot 2700 = 67{,}536{,}000\ £ \quad (29)$$
i.e. a reduction of £14,029,200
[1 mark]
This investment will benefit all parties in the system: The total generation cost will reduce (i.e. constraints cost will reduce), and total annual demand charges will also reduce.
However, still the total annual revenue of TransCo is higher than its total annuitized investment cost, signifying a degree of underinvestment, Therefore Borduria Regulator will prefer to reject this investment and instead suggest a greater investment such that neither underinvestment nor overinvestment occurs in the system.
[1 mark]